# Nonuniform Coverage and the Terminal-Cloud Staircase

Christopher D. Long

October 11, 2026

**Abstract**

For independent Poisson read counts with a common recovery threshold, we minimize the expected number of unrecovered types under a fixed total read intensity. At threshold one, full uniformity is uniquely optimal. At higher thresholds, every minimizer is uniform on its positive support. We solve the remaining discrete optimization: the optimal support grows through an exact nondecreasing staircase with unique adjacent crossings and no skipped support sizes. A two-point collapse argument gives the optimizer classification, while the elasticity of a Poisson tail orders the phase boundaries. We also give concentration bounds, fixed-read transfer, and conservative guarantees under calibration uncertainty and random sampling environments. These bounds concern specified allocations; the expected-defect objective must be distinguished from the probability of recovering a prescribed number of types.

## 1 Introduction

When recovering a type requires multiple observations, spreading a small read budget over every type can be inefficient. The expected number of unrecovered types can instead be reduced by leaving some types inactive and allocating equally among the rest. We prove this statement and completely determine how the optimal support changes with the budget.

The central results are Theorems 2.1 and 2.2. The first reduces a continuous simplex optimization to equal-positive-coordinate allocations. The second gives a unique crossing between each adjacent pair of supports, orders all these crossings, and localizes the optimum to the two integers surrounding a single scale $M/\rho_t$.

The coverage interpretation is motivated by DNA storage [3]. Nonuniform coupon collection, caching, and self-organizing search share a classical probabilistic framework [5]; stochastic first-copy optimization is treated in [2]. Multiple-copy computations for arbitrary sampling laws are developed in [1].

Our companion paper [6] optimizes the probability of recovering at least $k$ types and extends that optimizer theorem to all positive common Gamma shapes. For shapes greater than one and $k \geq 2$, it also proves global crossing uniqueness and consecutive MDS support phases. The two staircase theorems have independent proofs. An expected-defect minimizer need not maximize MDS success: a support smaller than $k$ cannot recover $k$ types at all. The two objectives can therefore have different optimizing supports and phase boundaries.

Sections 3 and 4 contain the extremal proofs and examples. The remaining sections give a compact coverage toolkit: exact independent-recovery probabilities, concentration, fixed-read

transfer, and calibration guarantees. These probability identities and bounds are separated from the optimizer classification.

## 2 Model and principal results

For an integer $n \geq 1$, write $[n] = \{1, \ldots, n\}$ and $\Delta_n = \{p \in [0,1]^n : \sum_i p_i = 1\}$. Fix $M > 0$ and an integer threshold $t \geq 1$. Independent Poisson counts $N_i \sim \text{Poisson}(Mp_i)$ recover type $i$ exactly when $N_i \geq t$. Put $\lambda_i = Mp_i$, and define

$$\phi_t(x) = 1 - e^{-x} \sum_{j=0}^{t-1} \frac{x^j}{j!}, \qquad \psi_t(x) = 1 - \phi_t(x), \qquad U_{M,t} = \sum_{i=1}^{n} \mathbf{1}_{\{N_i < t\}}.$$

The expected terminal low-count cloud is

$$\Lambda_t(M; p) = \mathbb{E}U_{M,t} = \sum_i \psi_t(Mp_i).$$

An allocation is *active-uniform* when all its positive coordinates are equal. At threshold one, strict convexity gives

$$\Lambda_1(M; p) = \sum_i e^{-Mp_i} \geq ne^{-M/n},$$

with equality only at $p_i = 1/n$. For $t \geq 2$,

$$\psi_t''(x) = \frac{e^{-x} x^{t-2}}{(t-1)!}(x - (t-1)),$$

so this Jensen argument cannot apply globally.

**Theorem 2.1** (Active-uniform minimizers for the expected terminal cloud)**.** *Fix $n \geq 2$, $M > 0$, and $t \geq 2$. For $p \in \Delta_n$, define*

$$\Lambda_t(M; p) = \sum_{i=1}^{n} \psi_t(Mp_i), \qquad \psi_t(\lambda) = e^{-\lambda} \sum_{r=0}^{t-1} \frac{\lambda^r}{r!}.$$

*For $q = 1, \ldots, n$, set*

$$\mathcal{L}_q(M) = (n - q) + q\psi_t(M/q).$$

*Then*

$$\boxed{\min_{p \in \Delta_n} \Lambda_t(M; p) = \min_{1 \leq q \leq n} \mathcal{L}_q(M).}$$

*Moreover, every global minimizer is, up to permutation, uniform on its positive support. More precisely, if $p^*$ minimizes $\Lambda_t(M; \cdot)$, then there exists $q \in \{1, \ldots, n\}$ such that exactly $q$ coordinates of $p^*$ are equal to $1/q$, and all remaining coordinates are zero. Conversely, if $q$ minimizes $\mathcal{L}_q(M)$, then every vector with exactly $q$ entries equal to $1/q$ and the remaining entries equal to zero is a global minimizer.*

**Theorem 2.2** (Exact support staircase for the expected terminal cloud)**.** *Fix $n \geq 2$, $M > 0$, and $t \geq 1$. Let*

$$\mathcal{L}_q(M) = (n - q) + q\psi_t(M/q), \qquad 1 \leq q \leq n.$$

*Equivalently, minimizing $\mathcal{L}_q(M)$ is the same as maximizing*

$$A_q(M) = q\phi_t(M/q).$$

*If $t = 1$, then $q = n$ is the unique minimizer of $\mathcal{L}_q(M)$ for every $M > 0$.*

*Assume now that $t \geq 2$. There is a unique number $\rho_t > 0$ satisfying*

$$\phi_t(\rho_t) = \rho_t \phi_t'(\rho_t).$$

*Equivalently,*

$$e^{\rho_t} = \sum_{r=0}^{t-1} \frac{\rho_t^r}{r!} + \frac{\rho_t^t}{(t-1)!}.$$

*For each $q = 1, \ldots, n-1$, there is a unique phase boundary*

$$\beta_q \in (q\rho_t, (q+1)\rho_t)$$

*defined by*

$$q\phi_t(\beta_q/q) = (q+1)\phi_t(\beta_q/(q+1)).$$

*Set $\beta_0 = 0$ and $\beta_n = +\infty$. Then*

$$\beta_0 < \beta_1 < \cdots < \beta_{n-1} < \beta_n,$$

*and*

$$\arg\min_{1 \leq r \leq n} \mathcal{L}_r(M) = \{q : \beta_{q-1} \leq M \leq \beta_q\}.$$

*Equivalently, away from the phase boundaries,*

$$q^*(M) = q \quad \Longleftrightarrow \quad \beta_{q-1} < M < \beta_q.$$

*At $M = \beta_q$, the only tie is between support sizes $q$ and $q+1$. Thus the optimal active support size is a nondecreasing staircase in $M$, and it changes by one coordinate at a time.*

The case $n = 1$ is immediate. For $n \geq 2$, the following proofs establish both the complete optimizer classification and the global phase diagram without invoking any MDS success theorem.

# 3 Proofs of the allocation and staircase theorems

## 3.1 Pairwise collapse

**Theorem 3.1** (Abstract two-point collapse principle)**.** *Let $f : [0, \infty) \to \mathbb{R}$ be continuous. Suppose that for every $S > 0$, the function*

$$h_S(a) = f(a) + f(S - a), \qquad 0 \leq a \leq S,$$

*attains its minimum only among the three points*

$$0, \qquad S/2, \qquad S.$$

*Then, for every $n \geq 1$ and $M > 0$,*

$$\min_{\lambda_i \geq 0,\ \sum_i \lambda_i = M} \sum_{i=1}^{n} f(\lambda_i) = \min_{1 \leq q \leq n} \left\{(n-q)f(0) + qf(M/q)\right\}.$$

*Moreover, every global minimizer is active-uniform: up to permutation,*

$$\lambda_1 = \cdots = \lambda_q = M/q, \qquad \lambda_{q+1} = \cdots = \lambda_n = 0$$

*for some* $q \in \{1, \ldots, n\}$.

*Proof.* The allocation simplex is compact and the objective is continuous, so a global minimizer exists. Let $\lambda^*$ be one. If two positive coordinates of $\lambda^*$ are unequal, say $\lambda_i^*, \lambda_j^* > 0$ and $\lambda_i^* \neq \lambda_j^*$, set $S = \lambda_i^* + \lambda_j^*$. Holding all other coordinates fixed, the contribution of this pair is $h_S(\lambda_i^*)$. Since $\lambda_i^*$ is neither 0, nor $S/2$, nor $S$, the two-point hypothesis gives a replacement of the pair by one of $(0, S)$, $(S/2, S/2)$, or $(S, 0)$ that strictly decreases the objective while preserving the total sum. This contradicts minimality. Hence all positive coordinates of $\lambda^*$ are equal. If there are $q$ positive coordinates, their common value is $M/q$, and the objective value is $(n-q)f(0) + qf(M/q)$. Taking the best $q$ gives the formula and the characterization of minimizers. □

**Lemma 3.2** (Two-point radial collapse)**.** *Fix* $t \geq 2$ *and* $S > 0$. *Define*

$$h_S(a) = \psi_t(a) + \psi_t(S - a), \qquad 0 \leq a \leq S.$$

*Then every minimum of* $h_S$ *on* $[0, S]$ *occurs at one of*

$$a = 0, \qquad a = S/2, \qquad a = S.$$

*Equivalently, if* $0 < a < S$ *and* $a \neq S/2$, *then there is a point*

$$b \in \{0, S/2, S\}$$

*such that*

$$h_S(b) < h_S(a).$$

*Proof.* We have

$$\psi_t'(\lambda) = -e^{-\lambda} \frac{\lambda^{t-1}}{(t-1)!}.$$

Set

$$g(\lambda) = e^{-\lambda} \frac{\lambda^{t-1}}{(t-1)!}, \qquad \lambda > 0.$$

The normalizing factor cancels in the logarithmic ratio below. On $(0, S)$,

$$h_S'(a) = -g(a) + g(S - a).$$

By the symmetry $h_S(a) = h_S(S - a)$, it is enough to study $0 < a \leq S/2$. Put

$$G(a) = \log \frac{g(a)}{g(S-a)} = (t-1) \log \frac{a}{S-a} + S - 2a.$$

Then

$$G'(a) = (t-1) \left( \frac{1}{a} + \frac{1}{S-a} \right) - 2 = (t-1) \frac{S}{a(S-a)} - 2,$$

and

$$G''(a) = -(t-1) \frac{S(S-2a)}{a^2(S-a)^2} < 0 \qquad (0 < a < S/2).$$

Thus $G$ is strictly concave on $(0, S/2)$. Moreover,

$$\lim_{a\downarrow 0} G(a) = -\infty, \qquad G(S/2) = 0.$$

The critical points of $h_S$ in $(0, S/2]$ are exactly the zeros of $G$. Since $G$ is strictly concave and the superlevel set $\{a : G(a) \geq 0\}$ is an interval containing $S/2$, either $G(a) < 0$ for all $0 < a < S/2$, or there is a unique $a_0 \in (0, S/2)$ such that

$$G(a) < 0 \quad (0 < a < a_0), \qquad G(a) > 0 \quad (a_0 < a < S/2).$$

In the first case, $g(a) < g(S-a)$, so $h_S'(a) > 0$ on $(0, S/2)$; hence the minimum on $[0, S/2]$ occurs at 0. In the second case, $h_S'(a) > 0$ on $(0, a_0)$ and $h_S'(a) < 0$ on $(a_0, S/2)$; hence $h_S$ first increases and then decreases, and the minimum on $[0, S/2]$ occurs at one of the endpoints 0 or $S/2$. By symmetry, the minimum on the whole interval $[0, S]$ occurs at one of $0, S/2, S$.

If $0 < a < S$ and $a \neq S/2$, then $a$ is not one of the three possible minimizers. Therefore its value is strictly larger than the minimum over the three-point set $\{0, S/2, S\}$, proving the final assertion. □

*Proof of Theorem 2.1.* Apply Theorem 3.1 to $f = \psi_t$, using Lemma 3.2. Since $\psi_t(0) = 1$, the active-uniform value with support $q$ is $(n-q) + q\psi_t(M/q)$. The change of variables $\lambda_i = Mp_i$ is a bijection between $\Delta_n$ and the allocation simplex. This proves the minimum formula and both directions of the equality classification. □

### 3.2 Ordering the support crossings

*Proof of Theorem 2.2.* Since

$$\mathcal{L}_q(M) = (n-q) + q\psi_t(M/q) = n - q\phi_t(M/q),$$

minimizing $\mathcal{L}_q(M)$ over $q$ is equivalent to maximizing

$$A_q(M) = q\phi_t(M/q).$$

For $x > 0$, define

$$a_t(x) = \frac{\phi_t(x)}{x}.$$

Then

$$A_q(M) = Ma_t(M/q).$$

For $t = 1$,

$$a_1(x) = \frac{1 - e^{-x}}{x}, \qquad a_1'(x) = \frac{(x+1)e^{-x} - 1}{x^2} < 0,$$

where the strict inequality is equivalent to $e^x > 1 + x$. Thus $a_1$ is strictly decreasing on $(0, \infty)$. Since $M/q$ is strictly decreasing as $q$ increases, $A_q(M) = Ma_1(M/q)$ is strictly increasing in $q$. Hence $q = n$ is the unique maximizer of $A_q(M)$, equivalently the unique minimizer of $\mathcal{L}_q(M)$.

Now assume $t \geq 2$. We first prove that $a_t$ has a single peak. Since

$$\phi_t'(x) = e^{-x}\frac{x^{t-1}}{(t-1)!},$$

set

$$s_t(x) = \frac{x\phi_t'(x)}{\phi_t(x)}.$$

Using

$$\phi_t(x) = e^{-x}\sum_{r=t}^{\infty}\frac{x^r}{r!},$$

we get

$$s_t(x) = \frac{x^t/(t-1)!}{\sum_{r=t}^{\infty} x^r/r!} = \frac{1}{(t-1)!\sum_{\ell=0}^{\infty} x^\ell/(t+\ell)!}.$$

The denominator is strictly increasing in $x$, so $s_t$ is strictly decreasing on $(0,\infty)$. Moreover,

$$\lim_{x\downarrow 0} s_t(x) = t > 1, \qquad \lim_{x\to\infty} s_t(x) = 0.$$

Hence there is a unique $\rho_t > 0$ such that $s_t(\rho_t) = 1$, which is exactly

$$\rho_t\phi_t'(\rho_t) = \phi_t(\rho_t).$$

Multiplying this identity by $e^{\rho_t}$ gives the displayed equivalent equation.

Furthermore,

$$\frac{d}{d\log x}\log a_t(x) = x\frac{d}{dx}\log\frac{\phi_t(x)}{x} = s_t(x) - 1.$$

Therefore $a_t$ is strictly increasing on $(0,\rho_t)$ and strictly decreasing on $(\rho_t,\infty)$.

We next prove that adjacent support sizes cross exactly once. Fix $0 < c < 1$, and define

$$D_c(x) = \log a_t(x) - \log a_t(cx).$$

Then

$$D_c'(x) = \frac{s_t(x)-1}{x} - c\frac{s_t(cx)-1}{cx} = \frac{s_t(x)-s_t(cx)}{x} < 0,$$

because $s_t$ is strictly decreasing and $cx < x$. Thus $D_c$ is strictly decreasing. Since $a_t$ is maximized at $\rho_t$,

$$D_c(\rho_t) > 0, \qquad D_c(\rho_t/c) < 0.$$

Therefore $D_c$ has a unique zero $\xi(c) \in (\rho_t, \rho_t/c)$.

For adjacent support sizes $q$ and $q+1$, take $c = q/(q+1)$. The equality

$$A_q(M) = A_{q+1}(M)$$

is equivalent to

$$a_t(M/q) = a_t(M/(q+1)).$$

Writing $x = M/q$, this becomes $a_t(x) = a_t(cx)$. Hence the unique crossing is

$$\beta_q = q\xi\left(\frac{q}{q+1}\right).$$

Since $\xi(q/(q+1)) \in (\rho_t, (q+1)\rho_t/q)$, we have

$$\beta_q \in (q\rho_t, (q+1)\rho_t).$$

In particular,

$$\beta_q < (q+1)\rho_t < \beta_{q+1},$$

so the phase boundaries are strictly ordered.

Finally, the sign of the adjacent difference is

$$A_q(M) > A_{q+1}(M) \quad \Longleftrightarrow \quad M < \beta_q,$$

and

$$A_q(M) < A_{q+1}(M) \quad \Longleftrightarrow \quad M > \beta_q.$$

If $\beta_{q-1} < M < \beta_q$, then

$$A_1(M) < \cdots < A_q(M) > A_{q+1}(M) > \cdots > A_n(M),$$

so $q$ is the unique maximizer of $A_r(M)$, equivalently the unique minimizer of $\mathcal{L}_r(M)$. If $M = \beta_q$, the same adjacent-sign argument gives the single tie

$$A_q(M) = A_{q+1}(M),$$

with strict inequalities on both sides. This proves the staircase description. □

## 4 Consequences and examples

**Corollary 4.1** (Full-support threshold for the expected terminal cloud)**.** *For $t = 1$, the globally uniform distribution $p_i = 1/n$ is the unique minimizer of $\Lambda_t(M; p)$ for every $M > 0$. For $t \geq 2$, with $\beta_{n-1}$ as in Theorem 2.2, the globally uniform distribution minimizes $\Lambda_t(M; p)$ if and only if*

$$M \geq \beta_{n-1}.$$

*It is the unique minimizer for $M > \beta_{n-1}$, while at $M = \beta_{n-1}$ it ties only with active-uniform laws of support size $n - 1$.*

**Corollary 4.2** (Localization of the optimal support)**.** *For $t \geq 2$, define the clipping map*

$$\kappa(r) = \min\{n, \max\{1, r\}\}.$$

*Every optimal support size $q \in \arg\min_{1 \leq r \leq n} \mathcal{L}_r(M)$ satisfies*

$$q \in \left\{ \kappa\left( \left\lfloor \frac{M}{\rho_t} \right\rfloor \right), \kappa\left( \left\lceil \frac{M}{\rho_t} \right\rceil \right) \right\}.$$

*Thus the exact support search is localized to the nearest integer support sizes around $M/\rho_t$, with only endpoint clipping by 1 and $n$.*

*Proof.* If $q$ is optimal, Theorem 2.2 gives $\beta_{q-1} \leq M \leq \beta_q$. For $1 < q < n$, the strict bounds on the phase boundaries imply

$$(q-1)\rho_t < \beta_{q-1} \leq M \leq \beta_q < (q+1)\rho_t.$$

Hence $q - 1 < M/\rho_t < q + 1$, so $q$ is one of the floor or ceiling of $M/\rho_t$. When $q = 1$ or $q = n$, the same conclusion follows after applying the clipping map to the one available finite boundary. □

**Example 4.3** (Low read depth and deliberate sacrifice)**.** Take $n = 2$, $t = 2$, and $M = 1$. The two candidate active support sizes are

$$\mathcal{L}_1(1) = 1 + \psi_2(1) = 1 + 2e^{-1} \approx 1.7358,$$

and

$$\mathcal{L}_2(1) = 2\psi_2(1/2) = 3e^{-1/2} \approx 1.8196.$$

Thus $q^* = 1$. The expected two-read terminal cloud is minimized by sampling only one strand and leaving the other unprotected. This illustrates why the higher-threshold objective is qualitatively different from the $t = 1$ case.

**Example 4.4** (Growth of the active support with read depth)**.** Let $n = 10$ and $t = 2$. The critical intensity $\rho_2$ is the unique solution of

$$e^{\rho_2} = 1 + \rho_2 + \rho_2^2,$$

so

$$\rho_2 \approx 1.79328.$$

The exact staircase theorem therefore predicts that the optimal support grows approximately like $M/\rho_2$, clipped at $n$. Direct evaluation of

$$\mathcal{L}_q(M) = 10 - q + q\psi_2(M/q)$$

gives the following representative minimizers:

| $M$ | $q^*$ | $\mathcal{L}_{q^*}(M)$ | $\mathcal{L}_{10}(M)$ |
|---|---|---|---|
| 5 | 3 | 8.5110 | 9.0980 |
| 10 | 6 | 7.0220 | 7.3576 |
| 20 | 10 | 4.0601 | 4.0601 |

At low read depth, the expected terminal cloud is minimized by protecting only a small active subset. As read depth increases, the optimal active support expands through adjacent support sizes, and eventually the full uniform distribution becomes optimal for this objective.

*Remark* 4.5 (Water-filling interpretation). Theorem 2.1 gives a precise sacrifice principle for the expected low-count cloud when $t \geq 2$. At low read depth it may be better, for this objective, to protect only an active subset of strands and leave the rest unprotected. This resembles classical water-filling in that an allocation problem collapses to an active set and an equalization rule on that active set, but the direction is reversed: here inactive coordinates are deliberately sacrificed because the hard threshold makes small intensities inefficient. Theorem 2.2 completes this picture: the active support size is selected by an exact staircase with adjacent phase transitions, and the support is concentrated near $M/\rho_t$.

## 5 Coverage probabilities and concentration

In this toolkit, $n \geq 1$ and $1 \leq k \leq n$ are integers, $p \in \Delta_n$ is fixed, and $M \geq 0$ is a finite budget. These statements evaluate or bound a specified allocation rather than classify an optimum. Read

counts take values in $\mathbb{N} = \{0, 1, 2, \ldots\}$. If a total $N \sim \text{Poisson}(M)$ of independent reads is assigned with probabilities $p$, the joint probability generating function is

$$\mathbb{E} \prod_i z_i^{N_i} = \exp\left( M \left( \sum_i p_i z_i - 1 \right) \right) = \prod_i \exp(M p_i (z_i - 1)).$$

Thus Poisson splitting gives independent counts of means $Mp_i$. At $M = 0$, all counts vanish. For hard thresholds, $\Lambda_t(0; p) = n$ for every $p$; the preceding optimizer and uniqueness theorems retain their positive-budget hypotheses.

**Proposition 5.1** (Independent recovery and the Poisson-binomial tail)**.** *For recovery curves $s_i : \mathbb{N} \to [0, 1]$, assume explicitly that, conditional on the* full *count vector, the reconstruction decisions are independent and type $i$ succeeds with probability $s_i(N_i)$. Then the recovery indicators are independent Bernoulli variables with parameters*

$$\theta_i(M) = \mathbb{E} s_i(N_i) = \sum_{r=0}^{\infty} s_i(r) e^{-Mp_i} \frac{(Mp_i)^r}{r!}.$$

*Recovery of at least $k$ types has probability*

$$P_{\text{succ}}(M) = \mathbb{P}\left( \sum_i B_i \geq k \right) = \sum_{j=k}^{n} [u^j] \prod_i (1 - \theta_i(M) + \theta_i(M) u).$$

*At $M = 0$, $\theta_i(0) = s_i(0)$, with the usual convention $0^0 = 1$ in the series. In particular, for $s_i(r) = \mathbf{1}_{\{r \geq t\}}$, one has $\theta_i(M) = \phi_t(Mp_i)$.*

*Proof.* For any recovery pattern, its conditional probability factors into functions of the individual counts. Taking expectations over the independent counts preserves this factorization and gives Bernoulli parameters $\theta_i$. Multiplication of their generating functions gives the coefficient formula. □

**Proposition 5.2** (Exact dynamic program)**.** *Let $d_i(j) = \mathbb{P}(\sum_{h=1}^{i} B_h = j)$, $0 \leq j < k$. With $d_0(0) = 1$, all other initial entries zero, and $d_i(-1) = 0$,*

$$d_i(j) = (1 - \theta_i) d_{i-1}(j) + \theta_i d_{i-1}(j - 1), \qquad P_{\text{succ}} = 1 - \sum_{j=0}^{k-1} d_n(j).$$

*Given the recovery probabilities $\theta_i$, this uses $O(nk)$ arithmetic operations and $O(k)$ memory. For arbitrary curves $s_i : \mathbb{N} \to [0, 1]$, $P_{\text{succ}}(M)$ is continuous on $[0, \infty)$. If every $s_i$ is nondecreasing, it is also nondecreasing in $M$; bisection then applies once a feasible target bracket has been established.*

*Proof.* Condition on $B_i$ to obtain the recurrence, and retain only the previous row. On $0 \leq M \leq A < \infty$, the absolute value of the $r$-th series term for $\theta_i(M)$ is at most $A^r/r!$. The series therefore converges uniformly on every compact budget interval, without a monotonicity assumption on $s_i$. The tail is a polynomial in the $\theta_i$, so it is continuous. When the curves are nondecreasing, a Poisson-increment coupling and the same independent uniforms for the decisions show monotonicity. □

**Proposition 5.3** (Terminal cloud and concentration)**.** *Under the common hard threshold, $U_{M,t}$ is Poisson-binomial with*

$$\Lambda = \mathbb{E}U_{M,t} = \sum_i \psi_t(Mp_i), \qquad V = \mathrm{Var}(U_{M,t}) = \sum_i \psi_t(Mp_i)(1-\psi_t(Mp_i)).$$

*An $[n,k]$ MDS code succeeds exactly when $U_{M,t} \le n-k$. Thus the first failing defect count is*

$$r_{\mathrm{fail}} = n-k+1.$$

*For every $s>0$,*

$$\mathbb{P}(U_{M,t} \ge \Lambda + s), \quad \mathbb{P}(U_{M,t} \le \Lambda - s) \ \le \ \exp\left(-\frac{s^2}{2V+2s/3}\right).$$

*If $\Lambda < r_{\mathrm{fail}}$, then*

$$\mathbb{P}(\text{failure}) \le \exp\left(-\frac{(r_{\mathrm{fail}}-\Lambda)^2}{2V+2(r_{\mathrm{fail}}-\Lambda)/3}\right).$$

*For general independent recovery indicators, let $\mu = \sum_i \theta_i$ and $v = \sum_i \theta_i(1-\theta_i)$. If $\mu > k-1$, then*

$$\mathbb{P}\left(\sum_i B_i < k\right) \le \exp\left(-\frac{(\mu-k+1)^2}{2v+2(\mu-k+1)/3}\right).$$

*In particular, for $0<\delta<1$, success probability is at least $1-\delta$ whenever*

$$\mu - k + 1 \ge \sqrt{2v\log(1/\delta)} + \frac{2}{3}\log(1/\delta).$$

*For the hard-threshold model, $\mu = n-\Lambda$ and $v = V$, so the two lattice gaps agree:*

$$\mu - k + 1 = r_{\mathrm{fail}} - \Lambda.$$

*Proof.* The defect indicators are independent Bernoulli variables with parameters $\psi_t(Mp_i)$, giving the moment formulas and recovery event. Since failure is exactly $U_{M,t} \ge r_{\mathrm{fail}}$, apply Bernstein's upper-tail inequality with $s = r_{\mathrm{fail}} - \Lambda > 0$. For general recovery indicators, $\sum_i B_i < k$ means $\sum_i B_i \le k-1$, so Bernstein's lower-tail inequality applies with displacement $a = \mu - k + 1 > 0$. If $L = \log(1/\delta)$ and $a \ge \sqrt{2vL} + 2L/3$, direct expansion gives $a^2 \ge L(2v + 2a/3)$, proving the sufficient margin condition [4, Chapter 2]. □

*Remark* 5.4 (Expected cloud is not success probability)*.* Minimizing $\Lambda$ need not maximize $\mathbb{P}(U_{M,t} \le n-k)$. For $n=2, t=2, M=1$, the expected-cloud optimum has support one, whereas recovery of both types requires positive allocation to both. The companion paper [6] optimizes the binomial tail for common Gamma shapes and proves its support-phase diagram independently. Those phase boundaries belong to the success-probability objective and need not coincide with the expected-cloud boundaries. Concentration bounds above can be useful for a chosen law, but do not identify an exact success-probability optimizer.

## 6 Fixed-read transfer

With exactly $m \in \mathbb{N}$ reads, the counts are $C^{(m)} \sim \text{Multinomial}(m; p)$. For an increasing count event $A \subseteq \mathbb{N}^n$, put $P_m(A) = \mathbb{P}(C^{(m)} \in A)$ and $\Pi_\lambda(A) = \mathbb{P}(Z^{(\lambda)} \in A)$, where $Z_i^{(\lambda)} \sim \text{Poisson}(\lambda p_i)$ independently.

**Theorem 6.1** (De-Poissonization sandwich)**.** *For every integer $m \geq 0$ and $0 < \varepsilon < 1$,*

$$P_m(A) \geq \Pi_{(1-\varepsilon)m}(A) - e^{-m[-\log(1-\varepsilon)-\varepsilon]},$$
$$P_m(A) \leq \Pi_{(1+\varepsilon)m}(A) + e^{-m[\varepsilon-\log(1+\varepsilon)]}.$$

*Proof.* If $m = 0$, then $C^{(0)} = Z^{(0)} = 0$ almost surely and both exponential error terms equal one, so the inequalities hold directly. Assume now $m \geq 1$. Construct all $C^{(j)}$ from successive prefixes of one infinite sequence of independent reads. For an independent $T_\lambda \sim \text{Poisson}(\lambda)$, the vector $C^{(T_\lambda)}$ has the law of $Z^{(\lambda)}$. Monotonicity of $A$ gives

$$\Pi_{(1-\varepsilon)m}(A) \leq P_m(A) + \mathbb{P}(T_{(1-\varepsilon)m} > m),$$

$$P_m(A) \leq \Pi_{(1+\varepsilon)m}(A) + \mathbb{P}(T_{(1+\varepsilon)m} < m).$$

The Poisson exponential moment $\mathbb{E}e^{uT_\lambda} = e^{\lambda(e^u-1)}$, with $u = \log(m/\lambda)$, bounds the corresponding upper or lower tail by $\exp[-m\log(m/\lambda) + m - \lambda]$. Substitution of the two means proves the inequalities. □

**Corollary 6.2** (Monotone recovery rules)**.** *The sandwich applies to the hard-threshold MDS event $\{c : \#\{i : c_i \geq t\} \geq k\}$. More generally, it applies to an upward-closed family of recovered sets with nondecreasing recovery curves $s_i$, under the full-count conditional-independence assumption of Proposition 5.1.*

*Proof.* Use independent $U_i \sim \text{Unif}[0, 1]$ for the reconstruction randomness. For each fixed $u$,

$$c \longmapsto \mathbf{1}\{\{i : u_i \leq s_i(c_i)\} \text{ is recoverable}\}$$

is increasing. For each fixed $u$, the preceding prefix argument therefore gives the two sandwich inequalities for the corresponding deterministic count event, with error terms independent of $u$. Averaging those inequalities over the auxiliary uniforms $U_i$ gives the stated unconditional randomized-recovery bounds. □

*Remark* 6.3 (Scope of transfer). The inequalities hold for each fixed sampling law, not as an exact optimizer classification for multinomial counts. First-copy optimization is known [2]; higher-threshold fixed-read optimization requires a separate argument. Nor do arbitrary upward-closed recovery rules inherit the active-uniform theorem merely because their probabilities can be computed under independent recovery. If extra reads can reduce a recovery probability, the monotone-prefix argument does not apply.

# 7 Bias, random environments, and calibration

**Theorem 7.1** (Uniform benchmark and perturbative bias penalty)**.** *Work in the Poissonized fixed-p model with hard threshold $t = 1$. Then, for every probability vector $p = (p_1, \dots, p_n)$ and every $M > 0$,*

$$\Lambda_1(M; p) = \sum_{i=1}^{n} e^{-Mp_i} \geq ne^{-M/n}.$$

*Equality holds if and only if $p_i = 1/n$ for every $i$.*

*Moreover, suppose*

$$p_i = \frac{1 + \varepsilon_i}{n}, \qquad \sum_{i=1}^{n} \varepsilon_i = 0, \qquad \varepsilon_i \geq -1.$$

*Thus the expansion includes boundary sampling laws with $p_i = 0$. Let*

$$\alpha = \frac{M}{n}, \qquad \sigma_2 = \frac{1}{n} \sum_{i=1}^{n} \varepsilon_i^2,$$

$$\sigma_3 = \frac{1}{n} \sum_{i=1}^{n} |\varepsilon_i|^3, \qquad \delta_\infty = \max_i |\varepsilon_i|.$$

*Then*

$$\Lambda_1(M; p) = ne^{-\alpha} \left(1 + \frac{\alpha^2}{2}\sigma_2 + R\right),$$

*where*

$$|R| \leq \frac{\alpha^3}{6} e^{\alpha\delta_\infty} \sigma_3.$$

*Proof.* The function $f(x) = e^{-Mx}$ is strictly convex on $[0, \infty)$. Jensen's inequality gives

$$\frac{1}{n} \sum_{i=1}^{n} e^{-Mp_i} \geq e^{-M(\frac{1}{n}\sum_i p_i)} = e^{-M/n}.$$

Thus $\Lambda_1(M; p) \geq ne^{-M/n}$. Strict convexity gives equality if and only if all $p_i$ are equal, hence iff $p_i = 1/n$ for every $i$.

For the expansion, write $\alpha = M/n$. Then

$$e^{-Mp_i} = e^{-\alpha} e^{-\alpha\varepsilon_i},$$

so

$$\Lambda_1(M; p) = e^{-\alpha} \sum_{i=1}^{n} e^{-\alpha\varepsilon_i}.$$

Taylor's formula gives

$$e^{-x} = 1 - x + \frac{x^2}{2} + \rho(x), \qquad |\rho(x)| \leq \frac{|x|^3}{6} e^{|x|}.$$

Taking $x = \alpha\varepsilon_i$, summing over $i$, and using $\sum_i \varepsilon_i = 0$, gives

$$\sum_i e^{-\alpha\varepsilon_i} = n + \frac{\alpha^2}{2} \sum_i \varepsilon_i^2 + \sum_i \rho_i,$$

with

$$|\rho_i| \le \frac{\alpha^3}{6} e^{\alpha\delta_\infty} |\varepsilon_i|^3.$$

After factoring out $ne^{-\alpha}$, the stated expansion and remainder bound follow. □

**Proposition 7.2** (Conditional environments and a two-stage guarantee)**.** *Let $0 < \delta, \eta < 1$, let $M \ge 0$ be a fixed finite budget, and let $P = (P_1, \dots, P_n)$ be a random probability vector. Conditional on $P$, suppose counts are independent Poisson variables of means $MP_i$. Put $q_M(P) = \mathbb{P}(\text{success} \mid P)$. Then the Poisson-binomial identity and concentration bounds above apply conditionally, with parameters $\phi_t(MP_i)$. The unconditional success probability is $\mathbb{E}q_M(P)$. If*

$$\mathbb{P}(q_M(P) \ge 1 - \delta) \ge 1 - \eta,$$

*then success probability is at least $(1 - \eta)(1 - \delta)$.*

*Proof.* Condition on the entire environment, apply the fixed-law identities, and take expectation. On the displayed good event, the conditional success probability is at least $1 - \delta$; elsewhere it is nonnegative. □

**Proposition 7.3** (Annealed one-read bias penalty)**.** *Let $M > 0$ and let $P = (P_1, \dots, P_n)$ be a random probability vector with $\mathbb{E}P_i = 1/n$ for every $i$. Then*

$$\mathbb{E}\Lambda_1(M; P) \ge ne^{-M/n},$$

*with equality exactly when $P_i = 1/n$ for every $i$ almost surely. Write $P_i = (1 + \varepsilon_i)/n$, $\alpha = M/n$, and suppose $|\varepsilon_i| \le L$ almost surely. Then*

$$\mathbb{E}\Lambda_1(M; P) = ne^{-\alpha}\left[1 + \frac{\alpha^2}{2n}\sum_i \mathbb{E}\varepsilon_i^2 + R_{\text{ann}}\right],$$

$$|R_{\text{ann}}| \le \frac{\alpha^3 e^{\alpha L}}{6n}\sum_i \mathbb{E}|\varepsilon_i|^3.$$

*Proof.* Apply strict Jensen to $e^{-MP_i}$ coordinatewise and sum. For the expansion, use $\sum_i \varepsilon_i = 0$ almost surely and $e^{-x} = 1 - x + x^2/2 + \rho(x)$, $|\rho(x)| \le |x|^3 e^{|x|}/6$, then take expectations. □

**Theorem 7.4** (Conservative calibration guarantee)**.** *For a fixed true law $p \in \Delta_n$ and a finite budget $M \ge 0$, let $0 \le \ell_i \le p_i$. Let $\underline{q}_\ell(M)$ be the probability of at least $k$ successes for independent Bernoulli variables with parameters $\phi_t(M\ell_i)$. Then the true Poissonized success probability $q_p(M)$ satisfies $q_p(M) \ge \underline{q}_\ell(M)$.*

*Let $0 < \delta, \eta < 1$. Suppose calibration data produce nonnegative simultaneous lower bounds $\widehat{\ell}_i$ such that $\mathbb{P}(\widehat{\ell}_i \le p_i \text{ for all } i) \ge 1 - \eta$. Choose a measurable depth rule $M : [0, \infty)^n \to [0, \infty)$. Conditional on calibration, subsequent counts are independent Poisson variables of means $M(\widehat{\ell})p_i$, generated using independent fresh randomness. If the selected finite depth satisfies*

$$\underline{q}_{\widehat{\ell}}(M(\widehat{\ell})) \ge 1 - \delta$$

*for every realized calibration vector, then the joint probability of success over calibration and subsequent sequencing is at least $(1 - \eta)(1 - \delta)$.*

*Proof.* Monotonicity of $\phi_t$ gives $\phi_t(M\ell_i) \le \phi_t(Mp_i)$. Coupling the corresponding Bernoulli indicators by the same independent uniform variables gives coordinatewise domination and hence the deterministic bound. Conditional on good calibration, that bound applies at the selected depth; averaging over calibration gives the two-stage probability guarantee. □

*Remark* 7.5 (Feasibility and decision rules). A calibration vector with fewer than $k$ positive lower bounds cannot yield a positive certified success probability, so feasibility of the depth-selection rule is a genuine assumption. Lower bounds are not renormalized, since renormalization could destroy coordinatewise conservativeness. A theorem valid pointwise in an environment does not permit a fixed allocation to be replaced silently by an adaptive allocation depending on the realized environment. In particular, $\max_p \mathbb{E}H(p, \mathcal{E}) \le \mathbb{E}\max_p H(p, \mathcal{E})$, and equality needs an additional argument.

## 8 Further questions and verification scope

The expected-cloud allocation and staircase problems above are solved for the common integer threshold under Poissonization. The companion MDS support phases are settled independently in [6]. Further questions concern exact higher-threshold fixed-read optimization, heterogeneous recovery costs, and risk-sensitive objectives such as a terminal-cloud moment-generating function. Independent reconstruction given the full counts, and monotonicity when transferring between read budgets, remain essential hypotheses.

The repository records statement-level dependencies and finite exact and numerical checks. These support reproducibility and future formalization but do not replace the analytic proofs; no proof-assistant certification is claimed. The original comprehensive source is preserved separately as an archive rather than included as a third active paper.

## References


[1] H. Abraham, I. Feldman, and E. Yaakobi, Analyzing Collection Strategies: A Computational Perspective on the Coupon Collector Problem, arXiv:2601.12351 (2026).

[2] E. Anceaume, Y. Busnel, E. Schulte-Geers, and B. Sericola, Optimization results for a generalized coupon collector problem, *J. Appl. Probab.* 53 (2016), 622–629; arXiv:1504.03878.

[3] D. Bar-Lev, O. Sabary, R. Gabrys, and E. Yaakobi, "Cover your bases: How to minimize the sequencing coverage in DNA storage systems," *IEEE Transactions on Information Theory*, vol. 71, no. 1, pp. 192–218, 2025. doi:10.1109/TIT.2024.3496587; arXiv:2305.05656.

[4] S. Boucheron, G. Lugosi, and P. Massart, *Concentration Inequalities: A Nonasymptotic Theory of Independence*. Oxford University Press, 2013.

[5] P. Flajolet, D. Gardy, and L. Thimonier, Birthday paradox, coupon collectors, caching algorithms and self-organizing search, *Discrete Applied Mathematics* 39 (1992), 207–229.

[6] C. D. Long, Optimal Allocations for Gamma Completion and MDS Recovery, companion manuscript (2026), repository source.